

Name: _____ Roll number: _____ House: _____

CS348 Computer Networks MidSem, 17 Sep 2025, 13:30-15:30, MaxMarks:30

General instructions

- Write only in the space provided. Answer briefly but crisply. Write neatly and clearly.
- You are allowed to refer to your own hand-written notes only.
- **All answers/calculations have to be (briefly) explained.** State any necessary assumptions.

Section A – Short Answer Questions [2 x 4 = 8 marks]

1. **Circuit vs Packet Switched:** Application-A generates constant length frames every 20ms, and requires a guaranteed upper bound on the delay each frame experiences while traversing the network from the sender to the receiver. Application-B also requires a guaranteed upper bound on the delay, but generates frames of different sizes every 20ms (e.g. video conferencing). Application-C generates frames of different sizes every 20ms, but does not require any guaranteed delay bounds.

a) A packet switched network is suited best for which of the above applications, and why?

b) A circuit switched network is suited best for which of the above applications, and why?

2. **Ethernet vs WiFi MAC:** Mention any one similarity between ethernet MAC and WiFi MAC. Also mention any one difference between the two.

3. **HDLC:** In HDLC, the delimiter bit-pattern used is 01111110. The sender stuffs a '0' after every 5 consecutive 1's in the data stream. For each of the following bit-patterns received at a HDLC link-layer receiver, in the middle of a frame, (a) remove any stuffed bits, and (b) indicate if there are any errors. In your answer, be sure to have a space between every 4 bits, for readability.

a) 1011 1101 1111 0001 1

b) 1100 1111 1000 0010 0

c) 0001 1111 1000 1100 1

d) 0110 1111 1110 0110 1

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4. **Supernetting:** An organization has 3 networks as follows: 190.168.20.0/24, 190.168.21.0/25, 190.168.21.128/25. How will the organization appear in the forwarding table of a “far away” router which has the same outgoing interface for all 3 networks? Explain briefly.

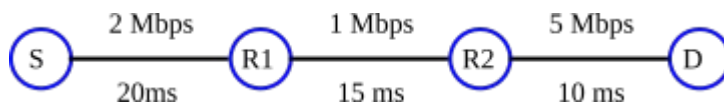
Section B – In-Depth Questions

5. **WiFi and Sliding Window Protocol:** WiFi link layer (in 802.11 a/b/g) uses the stop-and-wait protocol. And as per the MAC protocol, an ACK is sent *exactly* after SIFS, on successful reception of a data frame.

a) **[1 mark]** How many *bits* are *required* for the sequence number in the data frames? Why?

b) **[3 marks]** How many *bits* are *required* for the sequence number in the ACK frames? Why?
Hint: pay attention to how many marks this carries

6. **Store-and-Forward:** Consider the following store-and-forward datagram switched network between S and D. The link speeds and the propagation delays at each hop are as shown. A packet P1 of 1250 bytes is sent from S to D, followed immediately by a packet P2 of the same size. There is no other traffic on the network. Ignore processing delays.



a) **[1 mark]** At what time does D finish receiving P1 ?

b) **[3 marks]** At what time does D finish receiving P2 ? *Hint: your answering approach matters*

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7. **IP Forwarding [1x4=4 marks]:** Shown below is a forwarding table at a router in the public Internet (values are in hexadecimal, for easy conversion to binary). Explain your answers below briefly. *Hint:* Pay attention that the router is on the *public* Internet.

Destination network	Next-hop
0.0.0.0/0	N1
80.0.0.0/2	N2
83.F0.0.0/12	N3
83.FC.0.0/14	N4
83.FD.0.0/16	N5

- a) For a packet with destination address (in hexa) 83.FE.B0.0A, what is the next-hop chosen?

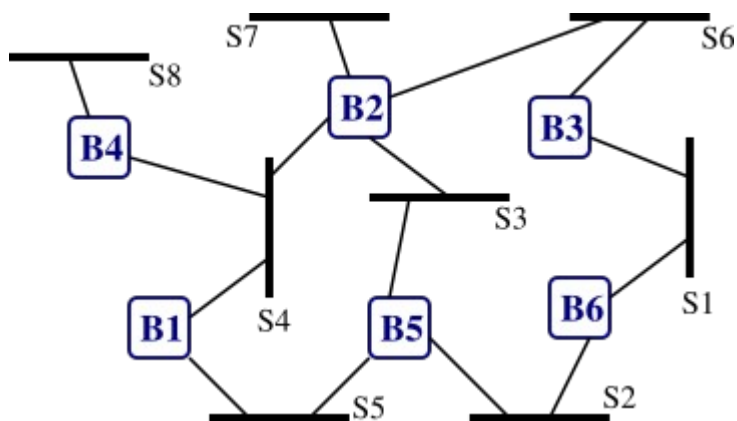
- b) For a packet with destination address (in hexa) B3.FE.B0.0A, what is the next-hop chosen?

- c) For a packet with destination address (in hexa) C3.FE.B0.0A, what is the next-hop chosen?

- d) For a packet with destination address (in hexa) 0A.B0.FE.83, what is the next-hop chosen?

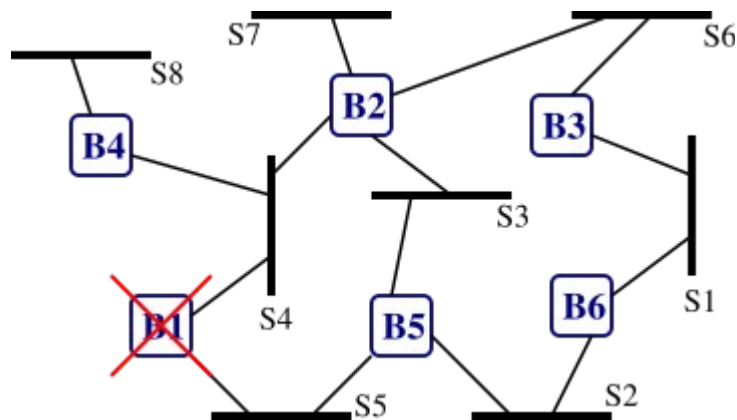
8. **Spanning Tree [5 marks]:** Answer the following with respect to an extended ethernet LAN with 6 bridges and 8 LAN segments as shown. Explain your answers briefly.

- a) Mark each port of each bridge as RP (root port), DP (designated port), or IN (inactive), after the spanning tree protocol has converged. Mark in *pencil* and *in the diagram itself* below.



- b) Repeat the above exercise after bridge B1 fails, and the spanning tree protocol has converged once again. Mark in *pencil* and *in the diagram itself* below (next page).

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9. **CSMA/CA [5 marks]:** Consider 3 WiFi nodes A, B, C in range of each other. The sequence of random numbers generated by each node is as follows: $A=[3,2,5]$, $B=[1,4,0]$, $C=[0,2,3]$. The *only* frames under consideration are as follows: A has 5 broadcast frames in its queue, B has 2 frames destined to C in its queue. RTS/CTS is turned *off* by configuration, and there are *no* channel errors. Ignore any propagation delays. Show the operation of the MAC protocol, with all relevant delays marked, until *successful* ACK of B's second frame. Answer in terms of a *neat timing diagram*. Note: it is best to draw in pencil. "Fold" the timeline if necessary.